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RESEARCH ARTICLE

JEZ-B MOLECULAR AND DEVELOPMENTAL EVOLUTION WILEY

# High regenerative capacity is a general feature within colonial dendrophylliid corals (Anthozoa, Scleractinia)

Bruna Louise Pereira Luz[1,2,3] | David John Miller[3] | Marcelo Visentini Kitahara[1,2,4]

[1]Coastal and Ocean Systems Graduate Program, Federal University of Paraná, Pontal do Sul, Pontal do Paraná, Paraná, Brazil

[2]Center for Marine Biology, University of São Paulo, Praia do Cabelo Gordo, São Sebastião, Brazil

[3]ARC Centre of Excellence for Coral Reef Studies and Centre for Tropical Bioinformatics and Molecular Biology, James Cook University, Townsville, Queensland, Australia

[4]Department of Marine Sciences, Federal University of São Paulo, Santos, São Paulo, Brazil

**Correspondence**
Bruna Louise Pereira Luz, ARC Centre of Excellence for Coral Reef Studies and Centre for Tropical Bioinformatics and Molecular Biology, James Cook University, Townsville, Queensland, 4811, Australia.
Email: bruna.pereiraluz1@jcu.edu.au.

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## Abstract

The regenerative capacity of cnidarians plays an essential role in the maintenance and restoration of coral reef ecosystems by allowing faster recovery from disturbances and more efficient small-scale dispersal. However, in the case of invasive species, this property may contribute to their dispersal and success in nonnative habitats. Given that four Indo-Pacific members of the coral genus *Tubastraea* have invaded the Atlantic, here we evaluated the ability of three of these species (*Tubastraea coccinea*, *Tubastraea diaphana*, and *Tubastraea micranthus*) to regenerate from fragments of undifferentiated coral tissue to fully functional polyps in response to differences in food supply and fragment size. For comparative purposes, another colonial dendrophylliid (*Dendrophyllia* sp.) was included in the analyses. All dendrophylliids displayed regenerative ability and high survival rates that were independent of whether or not food was supplied or fragment size. However, regeneration rates varied between species and were influenced by fragment size. Temporal expression of key genes of the regenerative process (Wnt and FGF) was profiled during whole-body regeneration of *T. coccinea*, suggesting a remarkable regenerative ability of *T. coccinea* that points to its potential use as a laboratory model for the investigation of regeneration in colonial calcified anthozoans.

KEYWORDS

dendrophylliidae, fragmentation, invasive species, regeneration, scleractinia, *Tubastraea coccinea*

## 1 | INTRODUCTION

Regeneration is a somatic function that enables organisms to regrow missing tissue and structures. Although common in some metazoan lineages, regeneration capacity is a highly variable property across species and organ systems (Bely & Nyberg, 2010; Bely, 2010; Goss, 1992; Reichman, 1984; Sun et al., 2018). While a few vertebrates, such as salamanders, can fully regenerate appendages after amputation (Bely & Nyberg, 2010; Sun et al., 2018), many others, inclluding mammals, have strikingly limited regenerative abilities that are essentially restricted to a few systems (e.g., blood and liver), and scarring being their general response (Poss et al., 2003; Poss, 2010; Reichman, 1984). In contrast, some early branching eumetazoans, such as cnidarians (e.g., *Hydra vulgaris*), can bypass the scarring stage, going through a phase known as "wound healing" on the way to regenerating missing structures or even the entire organism (i.e., whole-body regeneration) from small body pieces or even from a few dissociated cells (Bosch, 2007; Kirillova et al., 2016; Morgan, 1898; Technau et al., 2000). Nevertheless, the evolutionary changes underlying the loss of regenerative capacity in most complex metazoans remain largely elusive (Sun et al., 2018).

Overall, the regeneration capacity of an organism is determined by its plasticity potential for histogenesis and morphogenesis

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(Carnevali & Bonasoro, 2001), which in different organisms are mediated by the combination of (1) recruitment of stem cells and/or de-differentiation of cells, (2) cell proliferation and migration, and (3) expression or re-expression of the developmental program in adult stage (Carnevali & Bonasoro, 2001; Cha & Falanga, 2007; Galliot & Ghila, 2010; Giangrande & Licciano, 2014; Lai & Aboobaker, 2018). These processes are affected by environmental and biological conditions (Bak & Steward-Van Es, 1980; Bossert et al., 2013; Hall et al., 2015; Henry & Hart, 2005; Highsmith, 1982; Meesters & Bak, 1993; Sabine et al., 2015), such as temperature (Burmester et al., 2017; Dias et al., 2018; Kramarsky-Winter & Loya, 2000; Luz et al., 2018), health state (Ruiz-Diaz et al., 2016), and metabolic state (e.g., starving condition—Burmester et al., 2018; Galliot et al., 2018; Luz et al., 2018; Tomczyk et al., 2017). Ultimately these processes reflect the balance between energy cost and available cellular resources against other "investments," such as reproduction and somatic growth (Henry & Hart, 2005; Kramarsky-Winter & Loya, 2000; Lai & Aboobaker, 2018; Poss, 2010; Sebestyén et al., 2018).

The remarkable regenerative capacity of cnidarians (e.g., Amiel et al., 2015; Bosch, 2007; Kirillova et al., 2016; Lizcano-Sandoval et al., 2018; Luz et al., 2018) is assured by high tissue plasticity and the presence of highly flexible multi- or pluripotent stem cells, such as interstitial cells and amoebocytes. These cells are also involved in the maintenance of a wide range of biological functions, and participate in homeostasis and reproduction (Galliot & Ghila, 2010; Sebestyén et al., 2018). Although sexual reproduction may limit the investment into somatic functions by increasing differentiation of stem cells into gametes, asexual offspring arise largely from stem cells (Sebestyén et al., 2018). Also, hydrozoans (e.g., *Hydra*) and sea anemones (e.g., *Nematostella*) can reactivate central developmental programs, such as the conserved Wnt, FGF, Notch, MAPK-CREB, and BMP/chordin signaling pathways (Chera et al., 2011; Dubuc et al., 2014; Mehta & Singh, 2019; Münder et al., 2013; Turwankar & Ghaskadbi, 2019). Amongst these, the Wnt and FGF signaling pathways in particular play key roles in axial-patterning, morphogenesis and cell movement, proliferation, and differentiation (Abnave & Ghigo, 2019; Chera et al., 2009; Dubuc et al., 2014; Lange et al., 2014; Lee & Miura, 2014; Maddaluno et al., 2017; Sun & Irvine, 2014; Vogg et al., 2019; Vriz et al., 2014). However, whereas there has been some progress in understanding regeneration in these non-calcifying cnidarians, the molecular and cellular mechanisms involved in structural (e.g., tentacles), as well as whole-body regeneration, are unknown in scleractinian corals.

Overall, colony or polyp (corallite) fragmentation combined with regeneration might act as an alternative mode of asexual reproduction in scleractinian corals, allowing small scale dispersal (e.g., Heyward & Collins, 1985; Highsmith, 1982; Jackson, 1986; Lirman, 2000; Luz et al., 2018; Smith & Hughes, 1999) and faster recovery from disturbances (Highsmith, 1982). Thus, the unravelling of the molecular bases of regenerative processes in scleractinian corals potentially has broader implications for management and could also contribute to coral reef restoration programs (Lizcano-Sandoval et al., 2018).

However, successful regeneration of scleractinian corals after fragmentation at the polyp level is apparently dependent on some morphological features (e.g., solitary or colonial form), and evolutionary history. For example, it has been suggested that the regenerative abilities of solitary fungiid corals may be less extensive than those of colonial dendrophylliids (Chadwick & Loya, 1990; Chadwick-Furman et al., 2000; Luz et al., 2018). Herein, a coral colony is defined as a composition of multiple individual polyps or modules (equivalent morphological units) physically interconnected by a tissue layer overlying the skeleton, and internally by the gastrovascular cavity. These polyps should have their gene repertoire derived from a common ancestral cell (i.e., zygote) through asexual reproduction (Boardman et al., 1973; Davidson et al., 2004; Vuorisalo & Tuomi, 1986), with acquired somatic mutations passed on to the new module, leading to genetic mosaicism (Hughes et al., 1992; Maier et al., 2012; Olsen et al., 2019; van Oppen et al., 2011).

The genus *Tubastraea* (Scleractinia, Dendrophylliidae) consists of six extant Indo-Pacific species, all of which are azooxanthellate and colonial (Cairns, 2001). Among them, three are invasive in the Atlantic Ocean, where in some cases, they outcompete native species (Creed et al., 2017; Luz & Kitahara, 2017; Miranda et al., 2016; Santos et al., 2013). In addition to their broad repertoire of reproductive strategies, the success of these Indo-Pacific species in colonizing new habitats in the Atlantic is attributed in part to their ability to regenerate (Carlos-Júnior et al., 2015; Glynn et al., 2008; Luz et al., 2018, 2020), including being able to undergo whole-body regeneration from small fragments (<0.5 $cm^2$) of the corallum (Luz et al., 2018). To better understand the regenerative capacity across the genus, here we evaluated the responses of *Tubastraea coccinea*, *T. micranthus*, and *T. diaphana* to fragmentation, and also include another colonial dendrophylliid (*Dendrophyllia* sp.) to assess regeneration capacity more generally within the family. Moreover, to advance our understanding of the molecular mechanisms underlying regeneration in Scleractinia, the temporal expression of components of the two evolutionary conserved pathways with key functions in axial-patterning and morphogenesis (Wnt and FGF) was investigated during whole-body regeneration of *T. coccinea*.

## 2 | MATERIALS AND METHODS

Two colonies of *T. diaphana*, *T. coccinea*, and *Dendrophyllia* sp. containing ~15 polyps each and two branches from two colonies of *T. micranthus* (~25 polyps each) were collected at Pelorus Island (18°33.177′ S and 146°29.301′ E), Great Barrier Reef Marine Park Zoning (MAP 7—Townsville), Australia. The colonies were acclimatized to the laboratory for 5 days. Corallites from each specimen were manually broken using forceps and 24 fragments without polyp structures, such as mouth or tentacles (i.e., only skeleton with undifferentiated tissue) were selected. As the regeneration process can be size-dependent (Forsman et al., 2015; Luz et al., 2018), these fragments were divided into two groups ($n = 12$) based on size (large and small). Considering corallite size variation between studied species, fragment groups were based on the following size ranges (mean ± *SD*): *Dendrophyllia* sp., 0.84±0.20 and 2.62±0.96 $cm^2$; *T.*

*coccinea*, 1.78 ± 0.88 and 1.94 ± 1.3 $cm^2$; *T. diaphana*, 0.66 ± 0.28 and 2.81 ± 1.26 $cm^2$; and *T. micranthus*, 1.0 ± 0.35 and 3.09 ± 1.09 $cm^2$.

Fragments were placed into a standard 12-well plate, totaling two plates per specimen (six small and six large fragments per plate). Each plate was maintained in a 2 L water-flow system aquarium (20 μm filtered seawater at 27°C). Potential effects of food availability in the success of whole-body regeneration and in the mode that it occurs were tested during the first 31 days after fragmentation by adding equal amounts of *Artemia* nauplii (hatched in the laboratory, $n = 70$) every other day to half of the fragments from each coral species (i.e., to one well-plate of each specimen). The other half of the fragments were unfed during the experiment. Remaining *Artemia* was removed after 1 h. To avoid potential effects of fluctuating environmental laboratory conditions (e.g., different water flux in the system), aquariums position in the water-flow system were changed on a random basis on the bench every day.

The observation of the regenerative process and the characterization of its landmarks were carried out daily following Luz et al. (2018) during the complete (31 days) experiment and also on Day 77th after fragmentation. Fragments were observed using the stereomicroscope Olympus SZ61, and both sides of the fragments were photographed using a ToupCam pro SciTech camera coupled to the stereomicroscope (ToupView software package), immediately after fragmentation and at the following stages (landmarks) of the regenerative process: (i) mouth appearance; (ii) full viable polyp formed (mouth and tentacles well-formed; i.e., with nematocyst batteries present); and (iii) skeleton first observed. The first fully viable polyp formed on the fragment marked the complete regeneration. A more comprehensive view of the regeneration process was achieved by examining images of four randomly-chosen fragments every day.

The area of tissue covering the skeleton of each fragment was estimated using the software ImageJ implemented on the open-source platform Fiji (Schindelin et al., 2012) on the first day of the experiment and at the time that a viable polyp was formed (i.e., complete regeneration).

## 2.1 | Statistical analyses

### 2.1.1 | Survival and regeneration

Survival analysis was performed to measure the survival rate across 31 and 77 days from fragmentation (i) between species, (ii) among fragments of differing size, and (iii) as a function of whether food was supplied or not. The fragments that did not survive displayed tissue necrosis, which later resulted in a "naked" skeleton. Survival analysis was also employed to assess the rate of complete regeneration events across 31 days between the same experimental treatments of survival rate.

The Meier-Kaplan curve technique (Kaplan & Meier, 1958) was applied to examine the correlation between experimental treatments and the percentage of surviving and complete regeneration over time. The long-rank non-parametric test was used to compare curves between treatments. Cox Proportional-Hazards Regression analysis (Fox & Weisberg, 2011) was used to evaluate the significance of each treatment. This method assumes that the hazard of an event (e.g., tissue necrosis) at a given time point can be modeled by the product of a baseline hazard function and a regression function of tested covariates (Sinnott & Cai, 2016). Pairwise comparisons (Peto & Peto, 1972) were undertaken for further examination of significant covariates. Cumulative Hazard Transformation was applied to assess the expected number of times for the observation of fully regenerated polyps over 31 days after fragmentation. All analyses were performed using the software R 3.5.1, applying the coxph, survfit and survdiff in the Survival package (Therneau & Lumley, 2016). Results were summarized and visualized using the Survminer package (Kassambara et al., 2018).

### 2.1.2 | Tissue retraction

The extent of tissue retraction (i.e., the variation in the area of tissue covering the skeleton between day one and the end of the experiment) for the development to a full viable polyp, a proxy for the relative metabolic cost of tissue reorganization in relation to time (days) was examined against the effect of the factors "species," "fragment size," and "food supply" using the non-parametric Kruskall-Wallis test (Haigh & Conover, 1981). The Bartlett test (Bartlett, 1937) and Dunn's Kruskal-Wallis Multiple Comparisons (Dunn, 1964) were used to assessing homoscedasticity and significant sources of variation among variable factors, respectively.

### 2.1.3 | Double-polyp regeneration

An atypical type of development whereby two polyps are initially formed has previously been observed during regeneration of *T. coccinea* and *T. tagusensis* fragments, and occurred more frequently with larger than with small fragments (Luz et al., 2018). Such alternative regeneration mode may provide a competitive advantage to fragments by enabling higher colony growth rates (Forsman et al., 2015). To evaluate whether the incidence of this alternative type of development differed between tested dendrophylliids and the effects of fragment size and/or food supply, two-tailed Z-tests were used.

## 2.2 | Reverse-transcription PCR (RT-PCR) analysis

To detect the expression of Wnt and FGF signaling pathways during *T. coccinea* whole-body regeneration, semi-quantitative RT-PCR was performed for the ligands Wnt3a and 7b, and FGF16 and 18 from RNA extractions of fragments at different time intervals post fragmentation (immediately following fragmentation [T1], and every 5 days after fragmentation [daf] during 20 days [T2-5]), and morphologically distinct regeneration phases (mouth formed [MO], and full viable polyp emerged [PO]). Predicted Wnt/FGF protein sequences (Wnt 3a and 7b, and FGF16 and 18) and other two "housekeeping" genes (Rho guanine nucleotide exchange factor 7 [ARHGEF7], and

**TABLE 1** Description of *Tubastraea coccinea* primers

| Name of the gene | Primer sequence | Tm[a] (°C) | Length of the fragment (bp) |
|---|---|---|---|
| ARHGEF7 | F: 5′CCGCAACAAAGAATGGGAGC3′<br>R: 5′ACACTTCAACTGCTCTGGGG3′ | 59.6<br>51.5 | 100 |
| CCDC74A | F: 5′ACTGGTTATGGCCCGAACAG3′<br>R: 5′CCGACTGCCTCATCTGAGTC3′ | 53.9<br>51.9 | 105 |
| Wnt3a | F: 5′GGCAAGTCAACCAACCGATG3′<br>R: 5′TTGCGGCAAAGTTCTCGTTG3′ | 55.2<br>56.3 | 196 |
| Wnt7b | F: 5′TTGTTGCGTCGGTTACAACG3′<br>R: 5′TCGTTCCAGCGATTGCCATT3′ | 54.5<br>57.7 | 231 |
| FGF16 | F: 5′CTTTCTGTTGGCCCGGATCT3′<br>R: 5′GTGTCTGAATGGCATGTGCG3′ | 55.5<br>54.6 | 173 |
| FGF18 | F: 5′GGAACGCAACGTTTATCGGA3′<br>R: 5′GCTGCTCAATATGATCACGG3′ | 55.0<br>50.6 | 273 |

[a]Annealing temperature.

Coiled-coil domain-containing protein 74A [CCDC74A]) used as expression level controls were retrieved from *Acropora tenuis* genome and also from preliminary transcripts of *T. coccinea* (data not shown). The later used as templates to design the oligonucleotide primers with the software Geneious Primer 2020.1.1 (https://www.geneious.com). Primer sequences, amplicon lengths, and annealing temperature (Tm) are given in Table 1.

Gene expression was analyzed from distinct fragments from the two colonies of *T. coccinea* tested here for regeneration capacity. The fragments were kept without food in the same water-flow system described above. Total RNA was extracted after pooling of three fragments at the same stage of regeneration (T1 to complete polyp formed) from each colony (total of 14 samples) using TRIzol Reagent (Invitrogen) following Chomczynski and Sacchi (Chomczynski & Sacchi, 1987), and dissolved in RNase free water. The purity of RNA extracts was verified using a spectrophotometer (Nanodrop ND-1000), and concentrations determined using an Invitrogen Qubit Fluorometer with RNA high sensitivity (HS) or broad range (BR) assays kits. Integrity and concentration were validated using the Agilent 2200 Tapestation with the Agilent RNA ScreenTape or High Sensitivity RNA ScreenTape systems (see Udvardi et al., 2008). cDNA was synthesized using SuperScript IV VILO Master Mix (Invitrogen) and PCR performed using Platinum SuperFi PCR Master Mix (Invitrogen) and following the parameters: an initial denaturation of 95°C for 3 min, thirty cycles of 95°C for 30 s, 55°C for 30 s, and 72°C for 30 s, and a final extension of 72°C for 5 min.

## 3 | RESULTS

### 3.1 | Survival and regeneration process

The dendrophylliid species tested displayed high regenerative capacity; all colonies survived and underwent structural regeneration (i.e., regrowth of missing structures along with the original axial polarity—Figure 1a) after their polyps were broken off in small fragments. Moreover, 93% ($n$ = 192) of these generated fragments survived, 90% undergoing whole-body regeneration (Figure 1b-f) as indicated by the formation of one or more fully viable polyps after fragmentation (Figure 2a-d). Even small fragments devoid of food were able to fully regenerate. Fragment survival did not differ significantly between species or treatments (i.e., fragment size and food provision). In total, only 14 of the 192 fragments (7%) perished (i.e., underwent tissue degradation), and another six (3%) survived but did not fully regenerate during the experiment: at the 77 days time point, these fragments had undifferentiated tissue (live tissue without polyp structures, such as mouth or tentacles) or had undergone only partial regeneration (only mouth was formed).

Following colony fragmentation, soft tissue immediately proximal to the injured portion of the colony initiated the reorganization process that likely involved wound healing. With the exception of the skeleton, all missing structures were fully restored within ~2 weeks (Figure 1a). Fragment regeneration was characterized by an initial phase of tissue retraction/reorganization, during which repair processes were also involved, followed by the emergence of new mouth rudiments (sometimes two or more in a single fragment). Subsequent development of these fragments consisted either of (a) tissue reorganization around new mouth rudiments leading to the formation of one small polyp with a developing tentacle crown per mouth (Figure 1b-f) or (b) reabsorption of one or more of these rudiments. In the case that more than one polyp was formed on the fragment, these were sometimes (but not always) interconnected by a common tissue region ("coenosarc") (Figure 2a). For most of the regenerated polyps, secretion of new skeletal material onto the original skeleton fragment (Figure 2b) could be observed before 77 days. For each species, the morphology of the regeneration process from fragments to viable polyps is shown in Figure 1b-f.

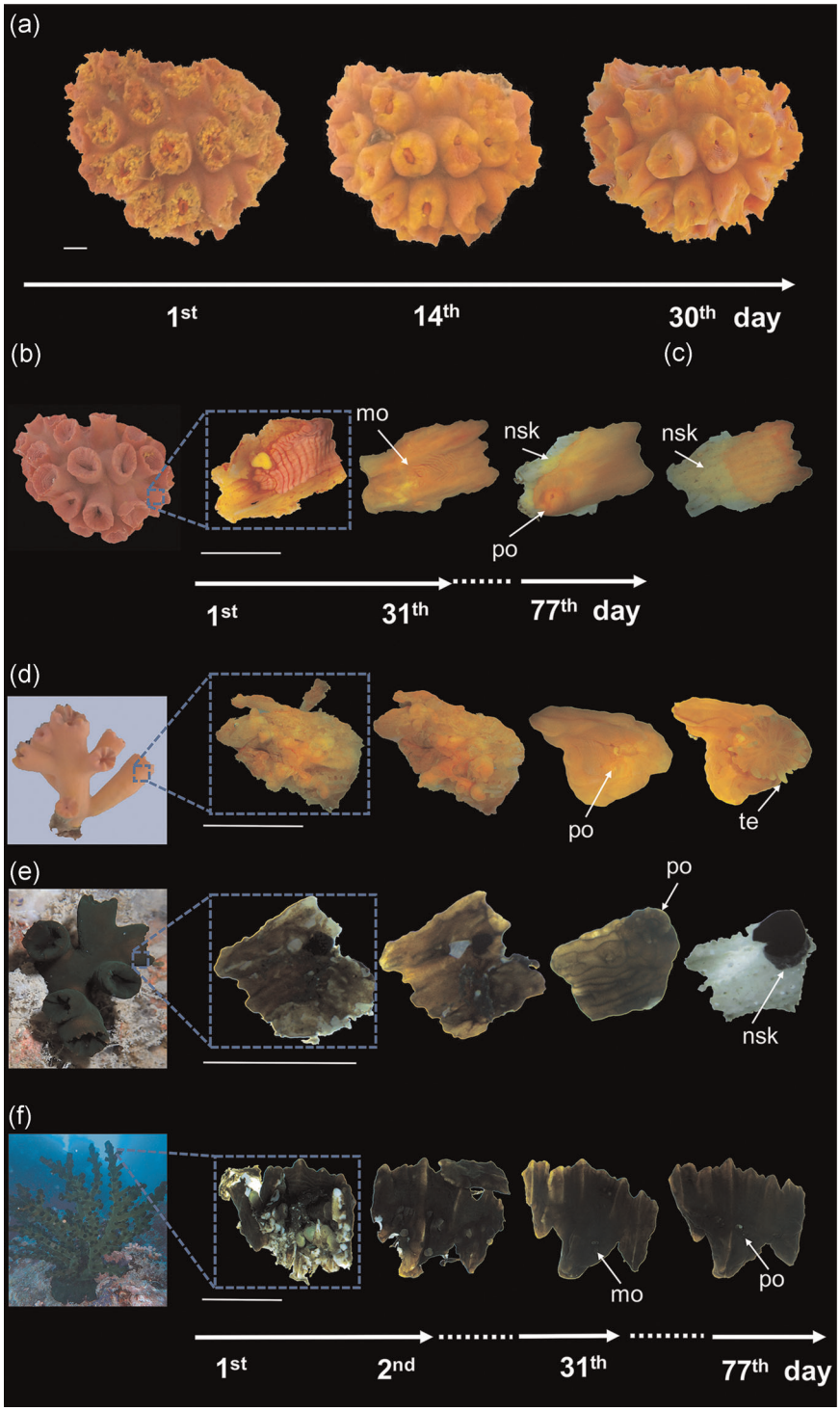


**FIGURE 1** Regeneration of dendrophylliid corals after fragmentation. (a) Adult colony of *Tubastraea coccinea* recovering from breakage. (b–f) Regeneration from undifferenced tissue on small fragments of (b, c)*Tubastraea coccinea*, (d) *Dendrophyllia* sp., (e) *Tubastraea diaphana*, and (f) *Tubastraea micranthus*. Arrows indicate: mo, mouth; po, polyp (i.e., fragment fully regenerated); nsk, polyp that has started to secrete skeleton; te, tentacle. Scale bars represent 5 mm [Color figure can be viewed at wileyonlinelibrary.com]

## 3.2 | Regeneration time

Regardless of size or food supply, 43% (83 of 192 fragments) of the dendrophylliid fragments had undergone complete regeneration by Day 31 (Figure 3). Of the species tested, regeneration occurred most rapidly in *T. diaphana*, followed by *Dendrophyllia* sp., while *T. coccinea* and *T. micranthus* did not differ significantly from each other in this respect (Figure 3a). Although only 10 of the 48 *T. coccinea* fragments (21%) had undergone complete regeneration before 31 days (Figure 3b), all of the fragments tested had regenerated fully by 77 days after fragmentation.

The firstcomplete and viable dendrophylliid polyps were noticeable 14 days after fragmentation, and by 27–31 days a large proportion (46 fragments) had completed regeneration. Other key events observed by the end of Day 31 were mouth rudiments ($n = 25$) and tentacle buds being present around the future oral end of the

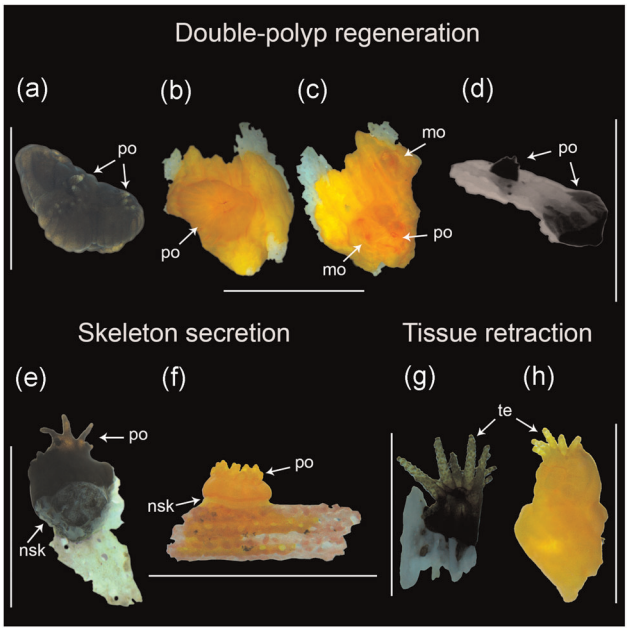


**FIGURE 2** Regeneration features of dendrophylliids corals after fragmentation. (a–d) Double-polyp regeneration resulting in interconnected polyps either in (a) the same side (*Tubastraea diaphana*) or (b, c) opposite site (*Tubastraea coccinea*) of the fragment, or with formation of (d) distinct polyps (*T. diaphana*). (e, f) Skeleton secretion from regenerated polyps of (e) *T. diaphana* and (f) *Dendrophyllia* sp. (g–h) Fragment regeneration with extensive tissue retraction in (g) *T. coccinea* or without significant tissue retraction in (h) *T. micranthus*. Arrows indicate: mo, mouth; po, polyp (i.e., fragment fully regenerated); nsk, polyp that has started to secrete skeleton; te, tentacle. Scale bars represent 5 mm [Color figure can be viewed at wileyonlinelibrary.com]

polyp ($n = 21$). Moreover, by Day 77, 66 additional fragments had undergone complete regeneration, totaling 172 fragments with at least one new polyp (~90%), of which 17% had secreted skeletons (Figure 4). Regeneration performance was observed to be size-dependent, as smaller fragments took around 5 days ($p < .001$) for mouth development and 3 days longer ($p = .23$) to regenerate to full polyp stage.

## 3.3 | Tissue retraction

The extent of tissue retraction (Figure 2g) in small fragments was significantly greater (involving approximately 2 × the area) compared to that from larger fragments ($\chi^2 = 5.1076$, $df = 1$, $p = .02$), and varied across species ($\chi^2 = 11.201$, $df = 3$, $p = .01$). In this sense, *T. coccinea* was more similar to *Dendrophyllia* sp. than to its congeners, both displaying more extensive tissue retraction (Table 2; Figure 5). In total, 14 of 192 *Tubastraea* fragments studied were able to regenerate without significant tissue retraction (Figure 2h); eight of these (~57%) were from *T. diaphana*, four (~30%) from *T. micranthus*, and two (~13%) from *T. coccinea*. It is important to note that the majority of these fragments (64%) were fed.

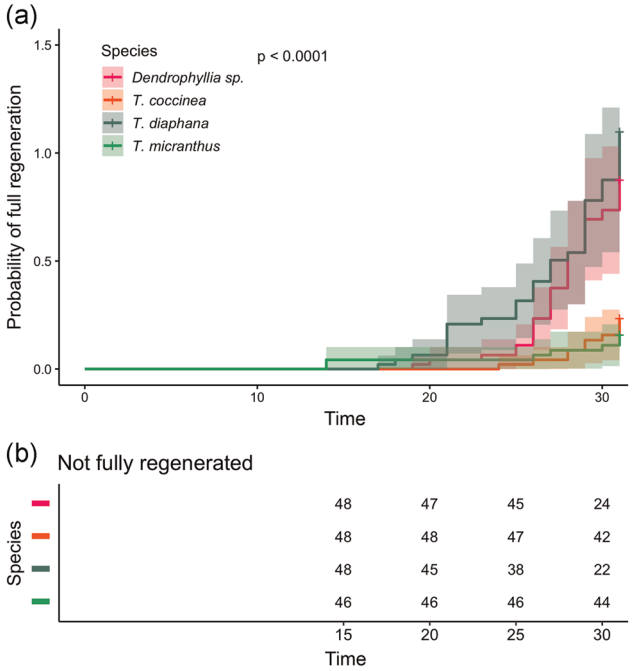


**FIGURE 3** (a) Kaplan–Meier estimates of cumulative hazard from complete regenerated polyps over 31 days after fragmentation; p indicates *p*-value measured by log-rank among survival curves (plot not exhibited). (b) Numbers of fragments of which recovery was incomplete at 15–30 days post-fragmentation

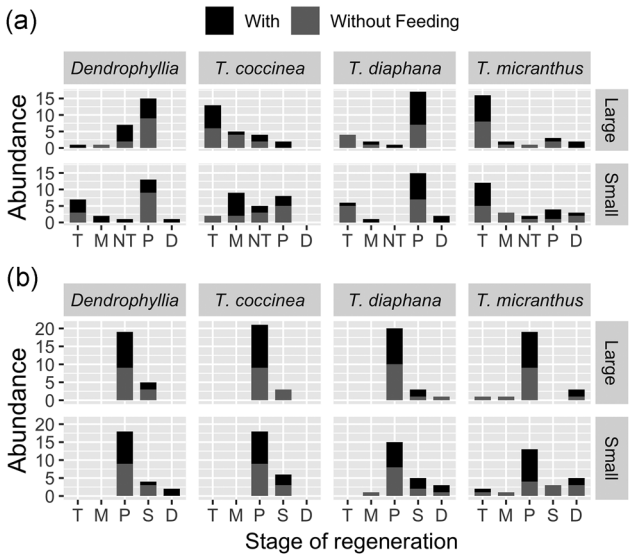


**FIGURE 4** The recovery process in fed and unfed coral fragments. The histograms show the abundance of dendrophylliid fragments (*Dendrophyllia* sp., *Tubastraea coccinea*, *Tubastraea diaphana*, and *Tubastraea micranthus*) at different stages of regeneration at (a) 31 and (b) 77 days after fragmentation sorted by fragment size (large and small). The proportion of fragments at each stage that were fed is represented in black, while unfed in grey. Letters indicate: T, tissue without visible structure; M, mouth visible; NT, tentacle primordia present; P, fully formed polyp (i.e., fragment complete regenerated); S, polyp in which skeleton secretion was observed; D, died. $n = 24$

**TABLE 2** Dunn test result comparing the percent of tissue retraction during whole-body regeneration in colonial dendrophylliid corals

| Species comparison | *Z* | *P* | *P-adj* |
|---|---|---|---|
| DE–TC | −0.622 | .534 | 0.534 |
| DE–TD | 2.364 | .018 | 0.109 |
| TC–TD | 2.320 | .020 | **0.041** |
| DE–TM | 2.201 | .028 | **0.042** |
| TC–TM | 2.352 | .019 | **0.056** |
| TD–TM | 0.764 | .445 | 0.534 |

The letters indicate: Z, *z*-score; P = *p* values; TC, *Tubastraea coccinea*; TD, *Tubastraea diaphana*; TM, *Tubastraea micranthus*; DE, *Dendrophyllia* sp. Numbers in bold indicate statistically significant p-adj (BH adjustment).

### 3.4 | Double-polyp regeneration

For a small proportion of the fragments, regeneration resulted in two polyps, which could be either distinct or interconnected by a common tissue. The polyps of the latter grew as a small colony (Figure 2a-c). Double-polyp regeneration was observed in nine fragments after 31 days, and an additional 11 fragments at the end of the experiment (77 days), representing ~11% and 6% of fully regenerated fragments at those time points, respectively. The frequency of occurrence of this phenomenon was independent of fragment size or feeding and did not differ between species.

### 3.5 | Wnt and FGF signaling pathways in whole-body regeneration

The Wnt and FGF signaling pathways were active in adult *T. coccinea* and during its whole-body regeneration. The Wnt7b and FGF16 and 18 ligands were expressed in adults and at all stages of the regeneration process investigated (following the fragmentation [T1] to full polyp formation [PO]). Semi-quantitative PCR indicated that FGF18 expression was relatively weak during the first two stages of regeneration (T1 and 2), and expression of Wnt3a was barely detectable at those time points (Figure 6). In contrast to the other ligands, two bands were seen after electrophoresis of Wnt3a PCR products (Figure 6), corresponding to either alternative splice variants or paralogous genes in scleractinians.

## 4 | DISCUSSION

Regeneration is an important feature in the life-history of scleractinian corals (Kramarsky-Winter & Loya, 1996, 2000; Tokuda et al., 2017), and plays an essential role in the maintenance of coral populations and reef ecosystems by enabling recovery following physical damage (Cameron & Edmunds, 2014; Forsman et al., 2015; Lirman, 2000). Fragmentation combined with regeneration is considered to constitute an alternative mode of asexual reproduction in corals (Heyward & Collins, 1985; Highsmith, 1982; Jackson, 1986; Lirman, 2000; Smith & Hughes, 1999). Therefore, the regenerative capacity of dendrophylliid corals belonging to the genera *Tubastraea* and *Dendrophyllia*, particularly their high survival rates and the ability to rapidly regenerate whole colonies from small fragments, indicates that regeneration is a general feature of their life-histories, and contributes to their ability to disperse and colonize new habitats.

Among the dendrophylliids examined (*T. coccinea*, *T. micranthus*, *T. diaphana* and *Dendrophyllia* sp.), *T. coccinea* was the most successful in recovering from fragmentation. Surprisingly, in this case, all of the fragments survived and fully regenerated, forming at least one new viable polyp by the end of the experiment (77 days). Similar regenerative capacity has previously been observed in colonies sampled from nonnative habitats (Luz et al., 2018). Therefore,

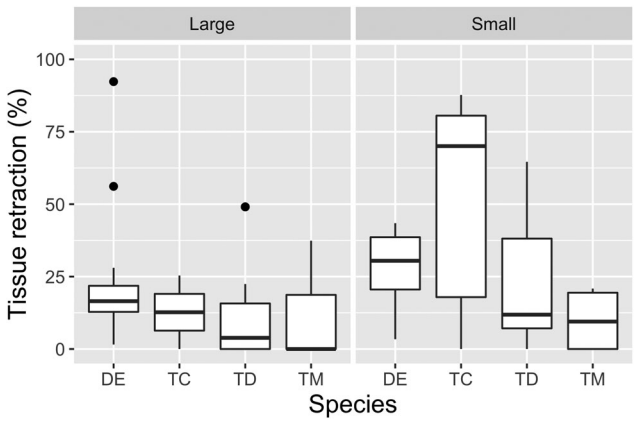


**FIGURE 5** The extent of tissue retraction during the regeneration process in fragments of *Dendrophyllia* sp. (DE), *Tubastraea coccinea* (TC), *Tubastraea diaphana* (TD), and *Tubastraea micranthus* (TM) according to fragment size. Differences between species (TC > TD and TM; $p = .01$) and between fragment size classes (small > large; $p < .001$) were statistically significant

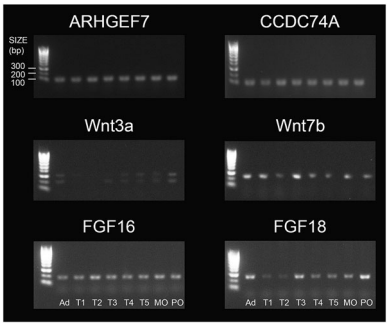


**FIGURE 6** Expression analysis of the "housekeeping" genes (ARHGEF7 and CCDC74A), and the ligands of Wnt (Wnt3a and 7b) and FGF (FGF16 and 18) signaling pathways during whole-body regeneration in *Tubastraea coccinea*. The letters indicate: Ad, adult; T1, immediately following fragmentation; T2-5, tissue remodeling from 5 to 20 daf; MO, mouth rudiment formed; PO, full polyp regenerated. The first lane from each gel corresponds to the Bioline HyperLadder 100-bp DNA marker

regeneration following fragmentation may be considered a strategy of *T. coccinea* to recover from damage as well as providing fast colonization rates in new invaded habitats.

*Tubastraea micranthus* and *T. diaphana* also occur in nonnative habitats and had high whole-body regeneration rates (73% and 90%, respectively). In contrast to *T. coccinea*, in those sites where they are invasive (i.e., Gulf of Mexico), both species remain restricted to artificial substrates, such oil and gas platforms (Sammarco et al., 2010, 2013). Nonetheless, their high survival and regeneration success rates suggest that they have the potential to further spread and establish populations in natural substrates.

For all of the dendrophylliids tested, the process of the whole-body regeneration was similar to that previously reported for *T. coccinea* and *T. tagusensis* from nonnative habitats, including tissue retraction and the formation of one or two small distinct but complete polyps (see Luz et al., 2018). Nevertheless, here we demonstrate that the process of skeleton deposition from regenerated polyps can occur earlier (before Day 77) than was previously thought, confirming that it may allow small scale dispersal at times outside of the sexual breeding season (Heyward & Collins, 1985; Highsmith, 1982; Jackson, 1986; Lirman, 2000; Smith & Hughes, 1999).

Regarding tissue reorganization, the extent of tissue retraction during the whole-body regeneration was greater in smaller fragments, being particularly pronounced in *T. coccinea* and *Dendrophyllia* sp. (Figure 5). Following Luz and collaborators (2018), food supply did not affect the rate of tissue retraction. However, fragments of *T. diaphana* and *T. micranthus* underwent complete regeneration with no apparent tissue resorption more frequently when fed. In *Hydra*, starvation conditions initiate the autophagy process, which results in decreased body size, but does not interfere with the ability to regenerate for weeks (Chera et al., 2009; Galliot et al., 2018; Tomczyk et al., 2017). In the case of scleractinian corals, starvation reduces protein concentration and organic matrix synthesis compared to feeding conditions (Ferrier-Pagès et al., 2003; Houlbrèque & Ferrier-Pagès, 2009), impacting healing processes and, consequently, regeneration in aposymbiotic corals, such as *Astrangia poculata* (Burmester et al., 2018). Therefore, the total energy available in the fragment may affect the regeneration process, and further studies should investigate the influence of prolonged starvation before colony fragmentation in addition to food supply during regeneration.

Taking into consideration the time frame of mouth and viable polyp observation, whole-body regeneration of smaller fragments occurred more slowly than for larger fragments. Such size-dependent performance has previously been documented for both sponges and corals and reflects intrinsic limitations imposed by the total energy and cellular resources available to an individual (Henry & Hart, 2005). Smaller fragments may contain incomplete cellular repertoires or insufficient numbers of stem cells to enable full regeneration. Therefore, their production costs impact in terms of both time and energy consumption. Together, although energy demand for tissue regeneration may vary between species, these results are consistent with the hypothesis that regeneration is more metabolically demanding for small fragments (Luz et al., 2018).

Unlike fragments, broken colonies underwent structural regeneration (i.e., regrowth of missing structures) along with the original axial polarity. This capacity is also observed in *Astrangia poculata* (Burmester et al., 2017, 2018; DeFilippo et al., 2016) and is comparable to the process of head regeneration of non-calcifying cnidarians (*Hydra*—Vogg et al., 2019; and *Nematostella*—Warner et al., 2018). Overall, injury healing was initiated immediately following damage, and the regeneration of missing structures was essentially completed within 14 days. Although the skeleton recovered much more slowly than tissue, this had no effect on polyp/colony survival. Such regeneration process implies that the healing process plays a decisive role in the survival of injured corals; securing body integrity allows the restoration of a barrier against potential pathogens, as well as limiting both tissue and/or cell loss and the potential for osmotic damage (Galliot & Ghila, 2010; Tiozzo & Copley, 2015). In fragments, wound healing must happen during tissue retraction/reorganization before the emergence of a new mouth. Molecular and cellular studies are needed to further improve the understanding of this process in corals, including identification of molecular markers during whole-body regeneration; in the case of other cnidarians (i.e., *Hydra*, *N. vectensis*, and *Calliactis polypus*), markers upregulated during wound healing include heat shock proteins and components of the MAPK signaling pathway (Dubuc et al., 2014; Stewart et al., 2017; Wenger et al., 2014, 2019).

Similarly to colony regeneration, detached adult polyps of *T. tagusensis* kept in laboratory may regenerate missing tissue and grow a complete polyp on each tissue-clad fenestra exposed at the broken surface (Luz, *personal observation*), as previously observed in *Porites* (Hubbard, 1973). The capacity to grow a second polyp instead of healing the distal portion indicates the potential of these single injured polyps to form a colony. This result is consistent with the double-polyp regeneration mode of dendrophylliid fragments, this property being independent of fragment size for the species tested here, as observed in *T. tagusensis* (Luz et al., 2018). Therefore, considering that regeneration is mediated by different combinations of cell proliferation, migration, and differentiation (Cha & Falanga, 2007; Lai & Aboobaker, 2018), the regeneration trajectory of an individual colony form may depend on the availability of cell types in the tissue remnant and cell plasticity, of which the latter probably varies interspecifically.

The expression of Wnt and FGF ligands during regeneration of *T. coccinea* implies that cell movement, morphogenesis and differentiation of cells and tissue occur during this process. Among the ligands studied (Wnt3a and 7b, and FGF16 and 18), only Wnt3a was extensively downregulated after fragmentation (T1), but active at 10 daf (T3). Moreover, at the first two time points of regeneration, FGF18 was only weakly expressed. These results raise the possibility of crosstalk between the FGF and Wnt pathways during *T. coccinea* regeneration, as occurs in many developmental contexts (e.g., Delgado & Torres, 2017; Galanternik et al., 2015). Additionally, the profile expression of Wnt3a suggests that the regeneration process in *T. coccinea* is similar to that observed during head regeneration after mid-gastric bisection in *Hydra* (Galliot & Chera, 2010; Gufler

et al., 2018; Vogg et al., 2019; Vriz et al., 2014), and after injury in *N. vectensis* (Dubuc et al., 2014). Overall, the regeneration of these cnidarians is initiated by cell death pathways, including apoptosis, resulting in removal of damaged cells and compensatory proliferation induced by releasing the ligand Wnt3 and subsequent activation of canonical Wnt signaling (Abnave & Ghigo, 2019; Chera et al., 2009; Dubuc et al., 2014; Lee & Miura, 2014; Sun & Irvine, 2014; Vriz et al., 2014). In *Hydra*, this pathway plays a key role in the axial patterning (Chera et al., 2009; Vogg et al., 2019), Wnt3 mediating the pre-patterning phase of head regeneration and the formation of the head organizer (Chera et al., 2009; Galliot & Chera, 2010; Guder et al., 2006; Hobmayer et al., 2000; Lengfeld et al., 2009; Petersen et al., 2015). Therefore, cell death pathways may also mediate the canonical Wnt pathway activation, compensatory proliferation, and regeneration in Scleractinia.

The relatively weak expression of FGF18 observed during the early regeneration stages in *T. coccinea* is comparable to that observed in *Nematostella* (Dubuc et al., 2014). Different from vertebrates, in *Nematostella* the expression of growth factors is delayed during wound healing (Dubuc et al., 2014; Werner & Grose, 2003). However, by contrast with *Nematostella*, FGF16 was expressed in all regeneration stages of *T. coccinea*. Moreover, in *Hydra*, FGF genes are expressed in interstitial cells (i-cells; Krishnapati & Ghaskadbi, 2013), mediating their differentiation into neurons and nematocysts (Lange et al., 2014), and also during hypostome and tentacle regeneration (Turwankar & Ghaskadbi, 2019). While i-cells have not been unequivocally identified in anthozoans, Richards and Rentzsch (2014) established that a comparable stem cell lineage is present; further studies are necessary to determine whether FGF signaling during early regeneration in *Tubastraea* plays similar roles as in *Hydra*.

Given the significance of regeneration in the life-history of corals and its potential to facilitate coral-reef restoration, it is important that molecular and cellular mechanisms underlying regeneration are better understood, particularly at a time when extreme weather events are becoming more common (Cheal et al., 2017; Hughes et al., 2018). By comparison to *Hydra*, the extent of knowledge about cell lineages and their fates in anthozoans is still limited, and cell lineage tracing would be an important step towards a better understanding of the regenerative processes of Scleractinia. Much of what is known about the molecular and cellular mechanism of regeneration in cnidarians is based on experiments carried out in hydrozoans (Gahan et al., 2016; Gufler et al., 2018; Holstein et al., 2003; Holstein, 2008; Kirillova et al., 2016; Passamaneck & Martindale, 2012; Petersen et al., 2015) or in actiniarians such *Nematostella vectensis* (Amiel et al., 2015; Darling et al., 2005; Warner et al., 2018). Despite their ecological and economic importance, there is a near-complete lack of information about the mechanisms that underlie regeneration in calcified anthozoans. Therefore, the striking regenerative capacity of dendrophylliids in general, and *T. coccinea* in particular, suggests that they may be useful laboratory models for further regeneration studies.

In addition, whole transcriptome analyses and investigation of the cellular processes that occur during regeneration of *T. coccinea*, including the mechanisms that mediate skeletogenesis may well provide information relevant to coral adaptation and reef restoration, which are particularly important in the present era of large-scale global reef degradation.

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## CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

## DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request. The peer review history for this article is available at https://publons.com/publon/10.1002/jez.b.23021

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